CS 105-01: Programming Languages

Instructor: R. Townsend
Time: MW 10:30-11:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/105-01

Description:
Principles and application of computer programming languages. Emphasizes ideas and techniques most relevant to practitioners, but includes foundations crucial for intellectual rigor: abstract syntax, lambda calculus, type systems, dynamic semantics. Case studies, reinforced by programming exercises. Grounding sufficient to read professional literature.

CS 112-01: Networks & Protocols

Instructor: F. Dogar
Time: TR 12:00-1:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/112-01

Description:
Design and implementation of computer communication networks, protocols, and applications, with an emphasis on the Internet protocol suite. Network architectures and programming interfaces. Data link, transport, and routing protocols. Congestion sources and remedies. Addressing and naming in local area and wide area networks. Network security and network management.

CS 115-01: Database Systems

Instructor: J. Bater
Time: TR 4:30-5:45p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/115-01

Description:
Fundamental concepts of database management systems. Topics include: data models (relational, object-oriented, and others); the SQL query language; implementation techniques of database management systems (storage and index structures, concurrency control, recovery, and query processing); management of unstructured and semistructured data; and scientific data collections.

CS 116-01: Introduction to Security

Instructor: M. Chow
Time: TR 4:30-5:45p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/116-01

Description:
A systems perspective on host-based and network-based computer security. Current vulnerabilities and measures for protecting hosts and networks. Firewalls and intrusion detection systems. Principles illustrated through hands-on programming projects.

CS 117-01: Internet-Scale Distributed Systems: Lessons from the World Wide Web

Instructor: N. Mendelsohn
Time: TR 4:30-5:45p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/117-01

Description:
Please note that this course was formerly numbered COMP 150-IDS.

The World Wide Web, one of the most important developments of our time, is a unique and in many ways innovative distributed system. This course will explore the design decisions that enabled the Web's success, and from those will derive important and sometimes surprising principles for the design of other modern distributed systems.

CS 118-01: Cloud Computing

Instructor: R. Sambasivan
Time: MW 4:30-5:45p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/118-01

Description:
Cloud computing fundamentals, including cloud architecture, scalability, elasticity, and metrics of cloud performance including service-level objectives (SLOs) and service-level agreements (SLAs). Cloud programming models and abstractions including Map/Reduce. Persistent storage mechanisms, including key/value stores and cold storage. Geo-distributed cloud systems. Cloud networking, including data center architecture, software defined networking, and middleboxes. Cloud security.

CS 121-01: Software Engineering

Instructor: J. Foster
Time: MW 3:00-4:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/121-01

Description:
Software engineering is an engineered discipline in which the aim is the production of software products, delivered on time and within a set budget, that satisfies the client's needs. It covers all aspects of software production ranging from the early stage of product concept to design and implementation to post-delivery maintenance. This course covers the major concepts and techniques of software engineering including understanding system requirements, finding appropriate engineering compromises, effective methods of design, coding, and testing, team software development, and the application of engineering tools so that students can prepare for their future careers as software engineers. The course will combine a strong technical focus with a project providing the opportunity to obtain hands-on experiences on entire phases and workflow of the software process.

CS 131-01: Artificial Intelligence

Instructor: To Be Announced
Time: MW 6:00p-7:15p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/131-01

Description:
History, theory, and computational methods of artificial intelligence. Basic concepts include representation of knowledge and computational methods for reasoning. One or two application areas will be studied, to be selected from expert systems, robotics, computer vision, natural language understanding, and planning.

CS 132-01: Computer Vision

Instructor: R. Shilkrot
Time: MW 6:00p-7:15p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/132-01

Description:
Introduction to low and intermediate levels of classic and modern Computer Vision. How to design algorithms that process visual scenes to automatically extract information. Fundamental principles and important applications of computer vision, including image formation, processing, detection and matching features, image segmentation, and multiple views. Basics of machine learning and deep learning for computer vision.

CS 133-01: Human-Robot Interaction

Instructor: E. Short
Time: MW 3:00-4:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/133-01

Description:
This course will provide an overview of the up and coming field of human-robot interaction (HRI) which is located squarely in the intersection of psychology, human factors engineering, computer science, and robotics. HRI has become a major research focus recently with the NSF's National Robotics Initiative and the push countries around the globe to develop robots for various societal tasks, from new flexible and adaptive robots for industrial manufacturing, to socially assistive robots for eldercare. In this course, we will examine this field from an interdisciplinary perspective, reading key papers in HRI that intersect computer science, robotics, cognitive and social psychology (since there is no suitable textbook yet, all reading materials will be made available). Students will give short presentations on HRI studies and designs and work in interdisciplinary groups on a term project which will require them to design and conduct an HRI study.

CS 134-01: Computational Models in Cognitive Science

Instructor: J. de Ruiter
Time: TR 9:00-10:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/134-01

Description:
Scientific logic of using computational models for testing theories in cognitive science. Connectionist and Bayesian models; agent-based simulation. Emphasis upon using models in combination with empirical data to test theories. Appropriate use and critical evaluation of computational modeling as found in scientific publications. Recommendations: COMP 10, 11, or some programming experience.

CS 135-01: Introduction to Machine Learning

Instructor: To Be Announced
Time: TR 9:00-10:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/135-01

Description:
An overview of methods whereby computers can learn from data or experience and make decisions accordingly. Topics include supervised learning, unsupervised learning, reinforcement learning, and knowledge extraction from large databases with applications to science, engineering, and medicine.

CS 136-01: Statistical Pattern Recognition

Instructor: M. Hughes
Time: TR 1:30-2:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/136-01

Description:
Statistical foundations and algorithms for machine learning with a focus on Bayesian modeling. Topics include: classification and regression problems, regularization, model selection, kernel methods, support vector machines, Gaussian processes, Graphical models.

CS 138-01: Reinforcement Learning

Instructor: J. Sinapov
Time: TR 10:30-11:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/138-01

Description:
"Reinforcement learning problems involve learning what to do — how to map situations to actions — so as to maximize a numerical reward signal." - Sutton and Barto ("Reinforcement Learning: An Introduction", course textbook)

This course will focus on agents that much learn, plan, and act in complex, non-deterministic environments. We will cover the main theory and approaches of Reinforcement Learning (RL), along with common software libraries and packages used to implement and test RL algorithms. The course is a graduate seminar with assigned readings and discussions. The content of the course will be guided in part by the interests of the students. It will cover at least the first several chapters of the course textbook. Beyond that, we will move to more advanced and recent readings from the field (e.g., transfer learning and deep RL) with an aim towards focusing on the practical successes and challenges relating to reinforcement learning.

There will be a programming component to the course in the form of a few short assignments and a final projects.

Approved as a category 2 elective in Data Science (analysis and interfaces).

CS 139-01: Ethics for AI, Robotics, and Human Robot Interaction

Instructor: M. Allen
Time: MW 9:00-10:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/139-01

Description:
This course will provide an overview of the ethical problems and challenges prompted by current and future technological advances in AI, robotics, and human-robot interaction. It will start by reviewing the philosophical foundations of the main ethical theories (virtue ethics, deontology, utilitarianism) and link them to different algorithmic approaches in artificial agents (rule-based, utility-based, behavior- based, etc.). Explicating and contrasting the assumptions underlying each algorithmic approach (e.g., policy-based decision-making vs. rule-based reasoning), functional tradeoffs and implications for autonomous robots and AI systems will be discussed. The scope will then be widened to moral psychology and human-robot/human-technology interaction to move beyond individual autonomous systems into the realm of social interactions between humans and autonomous systems, discussing the societal implications of AI and robot technology. Social, economical, legal, and military ramifications will be considered, with the aim of exposing the unique challenges AI and robot technology pose for humanity, compared to other disruptive technologies, but also they unique opportunities these technologies enable for current and future generations.

CS 150-01: Theoretical CompSci Toolkit

Instructor: V. Podolskii
Time: TR 12:00-1:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/150-01

Description:
In this course, we will explore various methods and techniques that are useful in various areas of computation theory. We will see how combinatorial and probabilistic technique, polynomial approach, Fourier analysis and linear algebraic technique can be applied to solve various problems in theoretical computer science. We will illustrate the techniques by the examples in theory of algorithms, computational complexity, learning theory, property testing and other areas.

CS 150-02: Advanced Programming Languages

Instructor: G. Wei
Time: TR 4:30-5:45p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/150-02

Description:
This graduate course explores advanced topics in programming languages. The primary goal of this course is to prepare students to explore the literature and perform research in this field. By the end of the course, students will be able to understand and implement results from research papers, as well as develop their own systems.

Topics include, but are not limited to: operational semantics (small-step reduction semantics, abstract machines, natural semantics), type systems (polymorphism, subtyping, type inference, refinement types, ownership), effects (continuations, algebraic effects and handlers, type-and-effect systems), analysis & transformation (CPS/ANF transformation, multi-stage programming, partial evaluation, symbolic execution), etc. The range of topics remains flexible and will be driven by student interests.

CS 150-03: Quantum Computer Science

Instructor: S. Mehraban
Time: TR 1:30-2:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/150-03

Description:
This is an elementary-level introduction to quantum computing through the computer science lens. One of the goals of this course is to present a language of quantum mechanics that is accessible to students working in computer science and vice versa. Topics include Hilbert spaces, the Dirac notation, the Schrödinger equation, quantum circuits, quantum Fourier transforms, quantum algorithms, and physical realizations of quantum computers. Students from different areas of engineering and sciences such as computer science, physics, electrical engineering, mathematics, or chemistry who wish to learn about the computer science foundations of quantum computers can benefit from this class.

CS 150-04: Generative Models

Instructor: L. Liu
Time: MW 10:30-11:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/150-04

Description:
Generative models have achieved remarkable success across various applications, including Large Language Models (LLMs) and multimodal generative AI. This course introduces the probabilistic foundations of deep generative models and provides hands-on experience in coding these models. We begin with simple distributions, exploring different distribution forms and their sampling procedures. Then, we transition to generative models based on neural networks. While these models are highly flexible and capable of capturing complex distributions, they require specialized learning algorithms for effective training. This course covers a range of these algorithms, with a primary focus on sequential models, diffusion-based models, and flow matching—key techniques underpinning modern generative AI. By the end of the course, students will develop a solid theoretical understanding of generative models and gain practical experience in implementing them.

CS 150-05: AI Foundations

Instructor: M. Scheutz
Time: MW 3:00-4:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/150-05

Description:
This course will introduce basic mathematical and computational concepts required for studying more advanced artificial intelligence algorithms and architectures.

CS 150-07: Algorithmic Music Composition

Instructor: R. Townsend
Time: MW 3:00-4:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/150-07

Description:
This course will explore music composition methods that use a purely algorithmic approach: examples include running a monte carlo method for a melody, deriving a chord progression from a grammar, or generating a whole piece with a cellular automaton. Students will compose their own pieces of music using these algorithmic techniques, and implement their compositions as programs. This course does not cover software tools for music generation (e.g., garageband, audacity, SuperCollider) nor how computers generate sound fundamentally.

CS 150-08: Scientific Machine Learning

Instructor: A. Patra
Time: TBA
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/150-08

Description:
Scientific Machine Learning: (in reality the use of Machine Learning and AI for science investigations) SciMl is an integration of traditional scientific computing and machine learning, which combines mechanistic models with data-driven reasoning, presented as a unified set of abstractions and a high-performance implementation. Science usage places different demands on ML tools in terms data availability, constraints and questions of interest and requires new approaches which are often generalizing to other traditional applications. For instance, while traditional deep learning methodologies have had difficulties with scientific issues like stiffness, interpretability, and enforcing physical constraints, this blend with data driven analytics and numerical analysis and differential equations has evolved into a field of research with new methods, architectures, and algorithms which overcome these problems while adding the data-driven automatic learning features of modern deep learning. Many successes have already been found, with tools like physics-informed neural networks, universal differential equations, solvers for high dimensional PDEs, and neural surrogates showcasing how ML/AI can greatly improve scientific modeling practice and support decision making where science must be used e.g.

climate adaptation.

Course will also build in training on software performance engineering.

CS 151-01: Cybersecurity Clinic

Instructor: M. Chow
Time: M 5:30p-7:00p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/151-01

Description:
Students are placed in interdisciplinary teams to work on a cybersecurity project with non-profit organizations in the Medford, Somerville, Cambridge, and Greater Boston communities.

CS 152-07: Goal Planning and Recognition

Instructor: R. Mirsky
Time: TR 1:30-2:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/152-07

Description:
This course introduces concepts and techniques for recognizing behavior, intentions, goals, and plans.

It presents formal definitions for various recognition problems, as well as a set of algorithms and approaches to solve each of them.

CS 160-01: Algorithms

Instructor: K. Edwards
Time: MW 4:30-5:45p, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/160-01

Description:
Introduction to the study of algorithms. Strategies such as divide-and-conquer, greedy methods, and dynamic programming. Graph algorithms, sorting, searching, integer arithmetic, hashing, and NP-complete problems.

CS 163-01: Computational Geometry

Instructor: D. Souvaine
Time: MW 1:30-2:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/163-01

Description:
(Cross-listed as MATH 181.)Design and analysis of algorithms for geometric problems. Topics include proof of lower bounds, convex hulls, searching and point location, plane sweep and arrangements of lines, Voronoi diagrams, intersection problems, decomposition and partitioning, farthest-pairs and closest-pairs, rectilinear computational geometry.

CS 165-01: Cryptography

Instructor: M. Ando
Time: MW 3:00-4:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/165-01

Description:
Introduction to cryptography, using a mathematical approach to prove properties about cryptographic systems, from the first ciphers to quantum cryptography. How codes and ciphers work, and how they can be broken. Private Key (Symmetric) and Public Key (Asymmetric) cryptography. Cryptographic protocols using block ciphers, methods for key exchange, hashing, message authentication, and digital signatures. Cryptographic protocols regarding secret sharing and digital cash.

CS 166-01: Computational Systems Biology

Instructor: S. Hassoun
Time: MW 10:30-11:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/166-01

Description:
Computational modeling of complex biological systems to analyze their emergent properties and understand or predict system behavior. Molecular and protein modeling in the context of biochemical networks. Application of these models across a select set of applications such as metabolomics, gut microbiome analysis, modularity analysis, flux balance analysis, and biological discovery. Introductions to machine learning, linear systems, and statistical concepts that are commonly used in systems biology.

CS 170-01: Computation Theory

Instructor: L. Cowen
Time: TR 1:30-2:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/170-01

Description:
Models of computation: Turing machines, pushdown automata, and finite automata. Grammars and formal languages including context-free languages and regular sets. Important problems including the halting problem and language equivalence theorems.

CS 175-01: Computer Graphics

Instructor: R. Chang
Time: TR 12:00-1:15, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/175-01

Description:
This course explores the fundamentals of computer graphics, including representing digital images, 2D rasterization and anti-aliasing, 3D rendering via ray casting, ray tracing and radiosity, viewing transformations, 3D shape representation, and an introduction to modeling and computer animation. Assignments and projects require a good working knowledge of the C programming language.

CS 201-01: Cyber for Future Policymakers

Instructor: D. Lillethun
Time: TRF 10:30-11:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/201-01

Description:
Relevance of computer technologies to policy development. Internet architecture and basic networking, the Web, cloud architectures, cryptography, security and privacy, AI and machine learning, and open-source systems. Developing technologies, including quantum computing and post-quantum cryptography. A recitation and graduate-level assignments are required. Four credit hours. Prerequisites: Graduate standing in a discipline other than Computer Science, Data Science, Bioinformatics, Cognitive Science, or Human-Robot Interaction.

CS 202-01: How Systems Work

Instructor: M. Sheldon
Time: TRF 10:30-11:45, Room To Be Announced
Course Page: https://www.cs.tufts.edu/t/courses/description/fall2025/CS/202-01

Description:
Graduate version of COMP13. How computing systems work: bits, bytes, the representation of information, the CPU, assembly language, programming languages. Networking: including peering, packets, and the Internet. Algorithms and the fundamental limitations of computing. Prerequisites: Graduate standing in a discipline other than Computer Science, Data Science, Bioinformatics, Cognitive Science, or Human-Robot Interaction.